

The Brain & the Autonomic Nervous System

Where the control centers sit, and the two settings the body runs on.

BY THE END OF THIS PAGE YOU SHOULD BE ABLE TO

- Match each brain region to the job it controls
- Say why brainstem damage is more deadly than damage higher up
- Trace a reflex arc, and explain why it does not wait for the brain
- Compare the sympathetic and parasympathetic systems, organ by organ
- Name the chemicals of each side, and predict what a drug that blocks them will do

1 The map that matters

You do not need every fold of the brain. You need to know which part failed when a patient changes.

REGION	WHAT IT DOES	WHAT ITS LOSS LOOKS LIKE
Cerebral cortex	Thinking, movement you choose, sensation, speech	Weakness, numbness, speech and personality changes
Cerebellum	Coordination, balance, smooth movement	Clumsy and unsteady, but strength is normal
Hypothalamus	Temperature, hunger, thirst, hormones, set points	Fever, fluid problems, hormone problems
Thalamus	Relay station for almost all sensation	Widespread sensory problems
Brainstem	Breathing, heart rate, blood pressure, being awake	Stopped breathing, collapse, coma

Notice the pattern. Damage high up costs you *abilities*. Damage low down costs you *life*.

The brainstem is small and plain, and it runs the machinery that can never pause.

WHY THE NURSE CARES

The hypothalamus holds the set point for the loops in Module 1. A fever is not a broken thermostat. It is the thermostat set higher on purpose. That is why the patient shivers on the way up.

2 The reflex arc — a decision without the brain

Some responses cannot wait for the trip upstairs.

- 1 **A receptor** notices the stimulus. Heat, stretch, or pain.
- 2 **A sensory neuron** carries the signal into the spinal cord.
- 3 **The cord decides.** Sometimes across a single synapse.
- 4 **A motor neuron** carries the command straight back out.
- 5 **The muscle acts.** Only then does the brain find out.

You pull your hand off a hot pan before you feel the burn. The brain is told at the same time, but it is not asked.

Testing a reflex tests one small segment of cord and its nerves on their own. That is exactly why reflexes are so good at locating an injury.

3 Two settings, and both are always on

Drag the dial and watch the whole body change together.

THE AUTONOMIC DIAL

Heart rate	resting balance
Pupils	resting balance
Airways	resting balance
Digestion	resting balance
Salivation	resting balance
Skin blood flow	resting balance
Sweat glands	resting balance
Bladder	resting balance
Liver glucose	resting balance

Neither side is ever fully off. What changes is the **balance**.

Every organ shifts at once, because the same outflow reaches all of them. That is why a frightened patient has a fast heart, wide pupils, a dry mouth, and a quiet gut all at the same time. One switch, many organs.

THINK OF IT THIS WAY

It is not an on/off switch. It is a mixing desk with two sliders. Push one up and pull the other down, and you get the same result either way.

4 The chemistry of the two sides

Learn the chemicals, and you can predict what any autonomic drug will do.

	SYMPATHETIC	PARASYMPATHETIC
Chemical at the organ	Norepinephrine, mostly	Acetylcholine
Receptor there	Adrenergic: alpha and beta	Muscarinic
Backup in the blood	Epinephrine from the adrenal gland	None
Main nerve route	Out of the chest and lower back	The vagus nerve, and the sacral nerves

Now the useful part.

A **beta blocker** blocks the sympathetic effect on the heart. So heart rate and force fall.

An **anticholinergic** blocks the parasympathetic side everywhere. So the heart speeds up, the pupils widen, the mouth dries, and the gut slows.

Those side-effect lists are not memorization. They are one mechanism, read across several organs.

The adrenal gland is a modified sympathetic nerve ending. Instead of touching an organ, it dumps epinephrine into the blood. That is why a fright outlasts the moment that caused it. The nerve signal ends in a second, but the hormone circulates for minutes.

5 Protection, and why some drugs cannot get in

The brain is guarded by three layers, a fluid cushion, and a wall of cells.

The **meninges** wrap it. **Cerebrospinal fluid** floats it, so it does not crush under its own weight. And the **blood-brain barrier** controls what leaves the blood.

Capillaries elsewhere in the body are leaky on purpose. In the brain they are sealed shut.

That barrier keeps toxins out. It also keeps many useful drugs out. This is why a brain infection needs antibiotics chosen for their ability to cross it, and why brain drugs are hard to design.

6 Check yourself

Five questions. The wrong answers explain themselves, so they are worth reading.

1. A patient has full strength but is clumsy and unsteady. Which part of the brain is most likely damaged?

- a) Cerebellum
- b) Brainstem
- c) Hypothalamus
- d) Thalamus

2. Why do you pull your hand away before you feel the pain?

- a) Pain travels more slowly than every other signal
- b) The spinal cord completes the loop without waiting for the brain
- c) The brain blocks pain during emergencies
- d) Motor neurons are faster than sensory neurons

3. Which set of signs fits strong sympathetic activation?

- a) Slow heart, small pupils, active gut
- b) Fast heart, wide pupils, dry mouth, slow gut
- c) Fast heart, small pupils, lots of saliva
- d) Slow heart, open airways, active digestion

4. A drug blocks muscarinic receptors. What would you expect?

- a) More saliva
- b) A slower heart rate
- c) Dry mouth and a faster heart rate
- d) Smaller pupils

5. Why must an antibiotic for meningitis be chosen for its ability to cross the blood-brain barrier?

- a) Spinal fluid destroys most antibiotics
- b) Brain capillaries are sealed and do not leak drug into the tissue
- c) The meninges absorb the drug first
- d) Brain tissue has no blood supply

Answer key

1. **Cerebellum**

Correct. The cerebellum smooths movement without supplying the power for it.

2. **The spinal cord completes the loop without waiting for the brain**

Right. The cord acts, and the brain is told afterward.

3. **Fast heart, wide pupils, dry mouth, slow gut**

Correct. One outflow, many organs, all shifted the same way.

4. **Dry mouth and a faster heart rate**

Right. Take away the parasympathetic side and the sympathetic side is left unopposed.

5. **Brain capillaries are sealed and do not leak drug into the tissue**

Correct. Elsewhere capillaries leak. In the brain they are sealed on purpose.

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